PROJECT PRESENTATION | PHASE-I

Hardware IP Dispute Verification

Zero-overhead biometric watermarking with asymmetric verification using Pedersen commitments and RSA-2048



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Github Repository Link: https://github.com/shubhamsaurav15107-art/minor_project

The Problem: Hardware IP Piracy

IP cores are shared as RTL code → **A dishonest firm copies the design** → **Sold under their brand, no licence fee**

Why traditional watermarking falls short

Wastes chip resources

Extra gates and dummy circuits hold the watermark, so the chip becomes larger, slower and more power hungry.

The “Inspection Paradox”

The same secret key that embeds the mark is needed to prove ownership. Showing it in a dispute lets anyone remove the watermark.

Our Solution & Key Contributions

Biometric Watermark

Unique points from a fingerprint, face or retina scan become a digital signature that cannot be guessed.

Zero-Overhead Embedding

The signature decides how variables share registers during High Level Synthesis: no extra gates or wires.

Safe Public Verification

Private embedding, public checking. Auditors use the public key and commitment, never the secret.

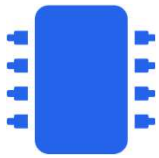
Working Web Platform

Vendor portal to register and sign designs, auditor portal to verify a suspect chip instantly.

Literature Review

17+

research papers on zero-knowledge proofs, biometrics and authentication (2012–2026)



ZKP authentication: Schnorr and SRP logins prove knowledge of a secret without sending it.

Biometrics + commitments: LocalVault rebuilt fingerprint secrets in 92% of attempts in under 1 second.

Blockchain & web apps: Healthcare, identity, e-voting and asset-proof systems keep data private yet verifiable.

Hardware & limits: FPGA speed-ups exist, but costs and interaction remain challenges.

Gap: no established practice for proving a hardware watermark in zero knowledge

Proposed Architecture

Vendor Environment

- Biometric scan creates a secret bitstream
- Generate RSA-2048 key pair
- Compute Pedersen commitment
- Sign netlist hash with private key

Register Server

- Validates the signature
- Stores public key and commitment
- Keeps a timestamped registry record in database.json

Verifier Dashboard

- Select registered design ID
- Input suspect register bitstream
- Enter commitment and blinding factor r
- Run verification, get verdict

Proposed Algorithm

Stage 1: Key & Watermark Generation

1. Derive secret bitstream from the vendor's biometric template
2. Create RSA-2048 key pair; register the public key
3. Lock the watermark with a Pedersen commitment
4. Hash the netlist with SHA-256 and sign it
5. Embed bits as register-conflict edges during HLS

Stage 2: Dispute Resolution

1. Structural analysis: recover the bitstream from the suspect's register allocation table
2. Cryptographic verification: check the signature and the commitment using public credentials
3. Verdict: infringement confirmed only if both match

Commitment: $C = h \cdot G + r \cdot H$ **Signature check: Verify_PuK(Hash || Bitstream, Signature)**

Technology Used

Component	Technology	Role
Backend	Python 3.9+	Server logic, cryptographic evaluation
Web server	Flask ≥ 2.0.0	API and routing
Asymmetric crypto	RSA, 2048-bit	Private / public key for verification
Hash engine	SHA-256	Netlist integrity and payload hashing
Data storage	Local JSON	Demo IP loaded from database.json
Frontend UI	HTML5 / CSS3 / JS	Interactive vendor and auditor portal

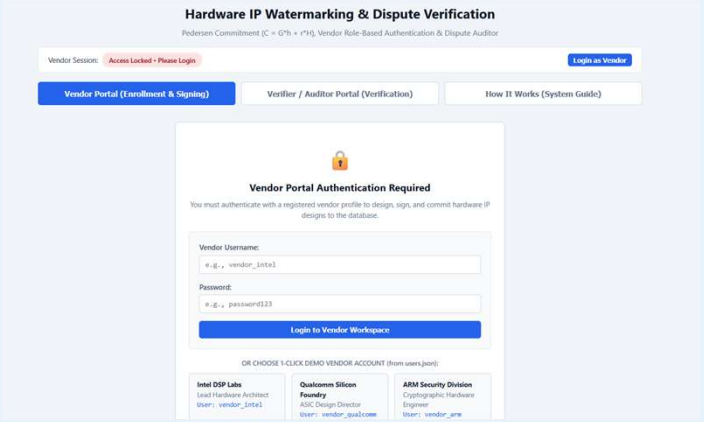


Figure: Outer look of the platform

Python handles the maths (commitment, signing); Flask connects it to the HTML/CSS/JS frontend.

Dispute Verdicts in Action

GENUINE
Clean, independently designed core

PIRATED
Watermark and signature both match

MODIFIED
Watermark found, netlist altered after signing

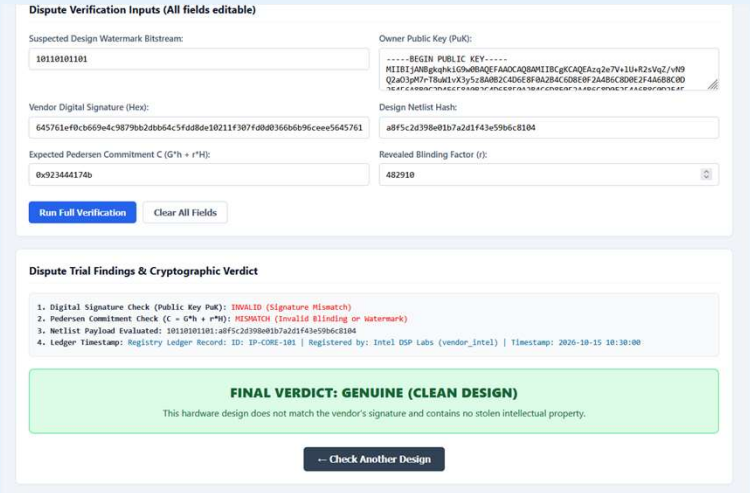


Figure: Verification of clean design

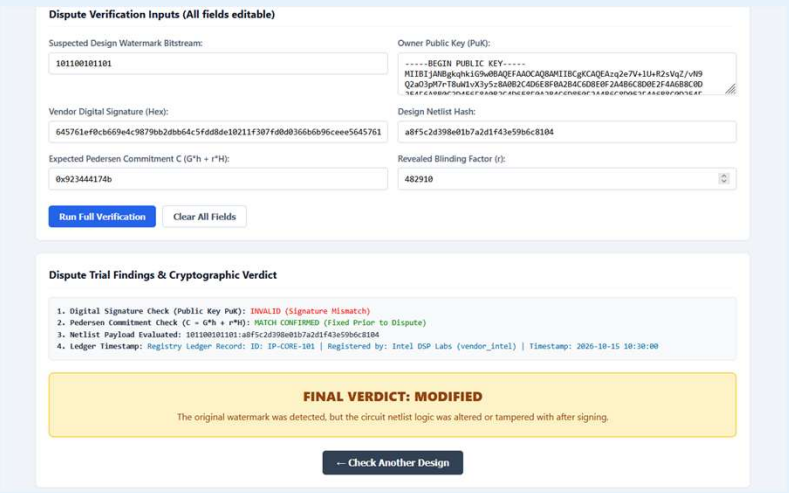


Figure: Verification of modified design

Results & Analysis

≈ 0

Hardware overhead. Speed and accuracy unchanged (at most 1–2 extra registers in some cases)

< 1 ms

Verification time, so the website gives an instant verdict

3

Real-world dispute scenarios tested successfully

Dispute testing

- **Stolen chip:** same watermark and hash, so declared pirated
- **Tampered chip:** hash changes, signature fails
- **Fake claim:** guessing the mark or key fails the maths

Key benefits

- Private key is never shared
- Pre-registered commitment proves the mark existed before the dispute
- Simple web dashboard for back-to-back checks

Conclusion & Future Scope

Conclusion

- Biometric watermark adds no area, delay or power cost
- Public key and Pedersen commitment verify ownership without exposing secrets
- Web platform detects pirated and tampered designs

Future scope

zk-SNARKs

Prove hidden rules are correct

Blockchain

Public notarization of hashes

Post-quantum

Swap RSA for ML-DSA

Netlist parser

Auto-read VHDL registers

Thank you

Reference

[1]A.Kumar, V.M. Shrimal and R.Rubbina, “Analysing security and efficiency in a qunatum resistant decentralised E-voting system using blockchain, homomorphic encryption and zero knowledge proofs,” in Proc. 2nd Int. Conf. Res. Methodologies Knowl. Manage., Artif. Intell. Telecommun. Eng. (RMKMATE), Chennai, India,2025,pp.1-5,doi: 10.1109/RMKMATE64874.2025.11042466.

[2]D.Naidu, B.Wanjari, R.Bhojwani, S.Suchak,R.Baser, and N.K.Ray, “Efficient smart contract for privarcy preserving authentication in blockchain using zero knowledge proof,” in Proc. OITS Int. Conf. Inf. Technol. (OCIT), Raipur, India, 2023, pp. 969-974, doi: 10.1109/OCT59427.2023.10430710.

[3]S.Mishra and R.K. Dwivedi, “Securing records in healthcare IoT using zero knowledge proof based blockchain,” in proc. 3rd Int. Conf. IoT,Commun. Authom. Technol.(ICICAT), Gorakhpur, India,2025, pp. 1-5,doi: 10.1109/ICICAT68430.2025.11414459.

[4]P. Baptista, B. Pacheco, F. Apolinario, J.Silveria, N.Escravana, and C. Riberiro, “Zero knowledge programming cerifications,” in Proc. IEEE Int. Conf. Distrib. Ledger Technol. (ICDLT), Pune, India, 2025, pp. 1-5, doi: 10.1109/ICDLT66400.2025.11466848.

[5]N. Datta, “Zero knowledge one time digital signature scheme,” in Proc. IEEE Int. Conf. Comput. Intell. Comput. Res., Coimbatore, India, 2012, pp. 1-4, doi:10.1109/ICCIC.2012.6510192.

[6]A. Abraham, M. Hiremath, and F.H. Pohrmen, “Privacy preserving authentication using zero knowledge proofs,” in Proc. IEEE 17th Int. Conf. Comput. Intell. Commun. Netw. (CICN), Goa, India, 2025, pp. 1003-1008, doi: 10.1109/CICN67655.2025.11368139.

[7]R. Akash, B. Bafna, A. Raghavan, R.T. Jose, P.B. Honnavalli, and I.Radhakrishnan, “LoaclVault: Zero-knowledge fingerprint authentication without biometric storage,” in Proc. 8th Int. Conf. Comput. Commun. Internet (ICCCI), Okayama, Japan, 2026,pp, 238-243, doi:10.1109/ICCCI70321.2026.11666520.

[8]V. Bobade, S. Ladwan, P. Waghmare, L.Ghai, P. Chaudhari, and C.puri, “A paradigm shift in digital trust: A cybersecurity analysis of zero- knowledge proofs,” in Proc. Int. Conf. Modern Sustain. Syst. (CMSS), Shah Alam, Malaysia, 2026, pp. 1573-1576, doi: 10.1109/CMSS69636.2026.11689222.

[9]P. Singhal and N. Joshi, “Zero-knowledge proofs for privacy- preserving agricultural yield verification: A blockchain-based incentive system for sustainable farming methods,” in Proc. 17th Int. Conf. Contemp. Comput. (IC3), Noida, India, 2025, pp. 1-6, doi:10.1109/IC366947.2025.11290217.

[10]X.Wang, “Research on the recovery strategy of federated learning model base on zero-knowledge proof,” in Proc. Int. Conf. Algorithm, Artif. Intell. Comput. Vis. (AAICV), London, U.K., 2025, pp. 194-198, doi: 10.1109/AAICV66571.2025.00042.

[11]S.A. Butt et al., “if-ZKP: Intel FPGA-based acceleration of zero knowledge proofs,” in Proc. IEEE 32nd Annu. INt .Symp. Field-Programmable Custom Comput. Mach. (FCCM), Orlando, FL, USA,2024, p.212, doi: 10.1109/FCCM60383.2024.00033.

[12]J. Kuriakose, P.S Sisodia, Amruth V, D.K. Shah, and S.More, “Comparative study of diverse zero-knowledge argument systems,” in Proc. Int. Conf. Data Mining adv. Comput. (SAPIENCE), Ernakulam, India,2016,pp. 284-293, odi: 10.1109//SAPIENCE.2016.7684134.

[13]V.K. Polisetti, A. Bhattacharjee, A. Gupta, and S.Vollala, “zkSHIELD- A web based cross chain zero knowledge asset proof,” in Proc.15th Int. Conf. Comput. Commun. Netw. Technol. (ICCCNT), Kamand, India,2024, pp. 1-5, doi: 10.1109/ICCCNT61001.2024.10726106.

[14]M. Chen, X. Qu, J. Cao, Z. Yang, and G.Sun, “Digital identity management based on blockchain and zero knowledge proof under the PIPL,” in Proc. 29th Int. Conf. Comput. Supported Cooperative Work Design (CSCWD),Fuzhou, China, 2026, pp. 340-345, doi: 10.1109/CSCWD68734.2026.11582607.

[15]V. Quach, R.Dantu, S. Talapuru, S.Zaman, and A. Pokharel, “ZCube: A zero-trust, zero-knowledge, and zero-memory platform for privacy and yet secured access,” in Proc. IEEE 6th Int. Conf. Trust, Privacy Secur. Intell. Syst. Appl. (TPS_ISA), Washington, DC, USA, 2024, pp. 166-175, doi: 10.1109/TPS-ISA62245.2024.00028.

[16]P.L. S.Kumari and A. Damodaram, “An alternative methodology for authentication and confidentality based on zero knowledge protocols using Diffe-Hellman key exchange,” in Proc. Int. Conf. Inf. Technol., Bhubneswar, India,2014, pp. 368-373, doi:10.1109/ICIT.2014.39.

[17]A. Pathak, T.Patil, S.pawar, P.Raut, and S. Khairnar, “Secure authentication using zero knowledge proof,” in Proc. Asian Conf. Innov. Technol. (ASIANCON), Pune, India, 2021, pp. 1-8, doi: 10.1109/ASIANCON51346.2021.9544807.

[18]M. Harikrishnan and K.V. Lakshmy, “Secure digital service payment using zero knowledge proof in distributed network,” in Proc. 5th Int. Conf. Adv. Comput. Commun. Syst. (ICACCS). Coimbatore, India, 2019, pp. 307-312, doi:10.1109/ICACCS.2019.8728462.

[19] M. Hao, C. Shang, S.Wang, W.Jiang, and J.Nie, “UAV- assisted zero knowledge model proof for generative AI: A multiagent deep reinforcement learning approach,” IEEE Internet Things J., vol.. 12, no. 10. pp. 13441-13454, May 2025, doi: 10.1109/JIOT.2025.3531914.

[20]P. Jagwani and S. Kaushik, “Defending location privacy using zero knowledge proof concept in location based services,” in Proc. IEEE 13th Int. Conf. Mobile Data Manage., Bengaluru, India, 2012, pp. 368-371, doi: 10.1109/MDM.2012.23.

[21] G.L.de Oliveira, F. A. Mendonca, and R.V. Ramos, “Zero knowledge protocol in a radio over fiber network,” in Proc. SBMO/IEEE MTT-S Int. Microw. Optoelectron. Conf.. (IMOC), Belem, Brazil, 2009, pp. 834-837, doi:10.1109/IMOC.2009.5427462.

[22]S. Liu, “Privacy protection revolution: Zero-knowledge proof,” in Proc. Int. Conf. Data Anal., Comput. Artif. Intell. (ICDACAI), Zakopane, Poland, 2022, pp.394-397, doi: 10.1109/ICDACAI57211.2022.00084

[23] A.M. Allam, I.I. Ibrahim, A.Ali, and A.E.H. Elsawy, “Effiecient zero-knowledge identification scheme with secret key exchange ,” in Proc. 46th Midwest Symp. Circuits Syst., Cairo, Egypt, 2003, vol. 1, pp. 516-519, doi: 10.1109/MWSCAS.2023.1572331.

[24]A. Sengupta and Rahul Chaurasia, “Retinal biometric for securing JPEG-Codec hardware IP core for CE systems”, IEEE Transactions on Computer Electronics, vol.69, pp. 441-457, Aug. 2023, doi: 10.1109/TCE.2023.3264669.